

KARAN SINGH

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AI Automation Developer & Data Analyst | B.Tech CSE specializing in Agentic Workflows, Computer Vision, and RAG-based systems. Proven track record in building high-impact solutions—from real-time pose estimation to hospitality automation—using Python, LangChain, and Power BI.

EXPERIENCE

WordPress Intern | Digital Word of Mouth | June 2025 – Aug 2025

- Web Development & AI Automation:** Engineered **41+ responsive WordPress pages** and integrated AI-driven social media automation tools to streamline content workflows.
- Performance Optimization:** Designed minimalistic, SEO-friendly layouts using **Bootstrap** and custom CSS to enhance site load speeds and user engagement.

Data Science Intern | Prodigy InfoTech | Jan 2025 – Feb 2025

- End-to-End ML Pipelines:** Developed 4 predictive models using **Python (Scikit-learn, Pandas)**, achieving a **15% accuracy boost** through advanced data preprocessing and feature engineering.
- Data Visualization & Insight:** Translated complex datasets into actionable business intelligence using **Matplotlib and Seaborn** to build interactive performance dashboards.

PROJECTS

HotelOps AI: Guest Management & Automation Suite | Present

- Intelligent Guest Lifecycle:** Designing an end-to-end guest management framework featuring **contactless digital check-in** and an AI-powered concierge to automate front-desk administrative workflows.
- Agentic Revenue Optimization:** Developing an **Agentic AI model** to identify real-time upselling opportunities, utilizing guest intent data to provide personalized service and room upgrade recommendations.

SalesGuru: AI Sales Mentor & Assistant | Feb 2026

- Orchestration & Reasoning:** Architected a sales mentorship framework using **LangChain** and **Groq**, utilizing advanced prompt chaining and high-speed inference to provide real-time strategic coaching deployed in Hugging Face.
- Knowledge Integration:** Developed the logic for context-aware responses, focusing on **RAG (Retrieval-Augmented Generation)** principles to synthesize sales methodologies into actionable user advice.

Pocket Coach: Posture & Reps Analyzer | [\(link\)](#) | Nov 2025 – Jan 2026

- Computer Vision Engineering:** Developed a real-time pose estimation engine using **MediaPipe** and **OpenCV** to track full-body landmarks and calculate precise joint angles for push-ups and squats.
- State-Machine Logic:** Implemented a robust state-machine for repetition counting and posture analysis, utilizing **Streamlit** for a dynamic UI that provides sub-second visual feedback on form quality.

Finance Dashboard | [\(link\)](#) | Oct 2025

- Business Intelligence:** Designed an interactive **Power BI** dashboard featuring advanced DAX-driven slicers for real-time Month-over-Month (MoM) growth and historical trend analysis.
- UX Optimization:** Integrated bookmark-based state management to facilitate rapid filter resets, ensuring a seamless user experience for high-density financial data segmentation.

SKILLS

AI & Machine Learning	Generative AI (LLMs, LangChain, Groq, RAG), Computer Vision (OpenCV, MediaPipe), Predictive Modeling (Scikit-learn)
Languages & Frameworks	Python (Expert), SQL, JavaScript, React, Django, Streamlit
Data & Cloud	Power BI (DAX), MongoDB, Oracle Cloud (OCI), Git, n8n Automation
Core Strengths	Full-Stack AI Development, Agentic Workflow Design, Rapid Prototyping

EDUCATION

B.Tech in Computer Science Engineering spec. AI & ML (pursuing)

Haridwar University, India | Aug 2023 – Aug 2027 | CGPA: 7.6

CERTIFICATIONS

Javascript with ReactJS - IIT Varanasi [\(link\)](#)

Machine Learning with Python - IBM [\(link\)](#)

Introduction to artificial Intelligence – IBM SkillsBuild [\(link\)](#)

Deep Learning with Computer Vision – IIT Madras (NPTEL)